

Boeing Certification Plans

1. Propose the criteria of its design
 2. Master Certification Plan: Submit detail plan in how different areas the FAA will directly oversee & which is dealt with for the ODA
 3. Perform qualitative & quantitative analyses, of detailed documents.
 4. FAA Evaluation (Phase 4): Conduct flight to prove safety, operational systems etc, all under actual flight conditions.
 5. Simulation, through testing structural & laboratory. In simulation they push the aircraft to its limit to fully test through simulation
 6. Double check
- Engineering/Manufacturing work instructions, is a specific step by step procedure provided - Blueprints

Essential Aircraft Anatomy

1. Fuselage
2. Wings
3. Cockpit
4. Propeller
5. Engine
6. Tail
7. Landing gear



Cessna 172 Main Structure

1. Skeletal of aircraft is aluminum ^{former} ~~structure~~, bulkheads, longerons, & stringers.
2. Landing gear material: flexible steel to withstand hard landing.
3. Wings: skeletal structure are ribs provides airfoil shape of wing. Spars run through entire length of wings; give strength & then stringers to support even more.
4. The skeletal system covered by aluminum sheets, which is called the air frame, not the engine & avionics; the whole structure.
5. Applies a green or yellow primer layer on skeletal system to prevent corrosion. It is zinc chromate or zinc phosphate.
6. Wing struts to reinforce the joint between the wings & the fuselage. Attach from main spar to lower fuselage. There is spar carry through channels that join the two wings together. Not as strong but that's what the wing struts are for.